Customer Churn Prediction & Recommendation System

Duration

3–4 Weeks

Difficulty

Advanced

Objective

Develop an intelligent Machine Learning platform that predicts customer churn, identifies the factors contributing to churn, segments customers based on behavior, and recommends personalized retention strategies. The application should provide predictive analytics through an interactive dashboard and REST APIs.

Technology Stack

- Python
- FastAPI
- Scikit-learn
- XGBoost / CatBoost / LightGBM
- Pandas
- NumPy
- Matplotlib / Plotly
- PostgreSQL or MongoDB
- React.js
- Docker
- Git & GitHub

Core Features

1. Authentication

- User Registration
- Login
- JWT Authentication
- Role-Based Access (Admin, Business Analyst)

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2. Customer Dataset Management

- Upload CSV/Excel Dataset
- Data Preview
- Missing Value Detection
- Duplicate Detection
- Dataset Validation

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3. Data Preprocessing

- Missing Value Imputation
- Feature Encoding
- Feature Scaling
- Outlier Detection
- Feature Engineering
- Feature Selection

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4. Exploratory Data Analysis (EDA)

Generate visual reports including:

- Customer Demographics
- Correlation Matrix
- Churn Distribution
- Feature Importance
- Customer Behavior Analysis

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5. Customer Segmentation

Cluster customers using:

- K-Means Clustering
- DBSCAN (Bonus)

Display customer groups and their characteristics.

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6. Churn Prediction

Train and compare multiple models:

- Logistic Regression
- Random Forest
- XGBoost
- LightGBM
- CatBoost

Predict whether a customer is likely to churn.

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7. Recommendation Engine

Based on model predictions, recommend retention actions such as:

- Special Discounts
- Loyalty Programs
- Personalized Offers
- Customer Follow-up Priority

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8. Model Evaluation

Evaluate using:

- Accuracy
- Precision
- Recall
- F1 Score

- ROC-AUC
- Confusion Matrix

Compare all models and select the best performer.

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9. Business Dashboard

Display:

- Total Customers
- Churn Rate
- High-Risk Customers
- Retention Rate
- Customer Segments
- Model Accuracy
- Prediction Trends

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10. REST API

Develop APIs for:

- Dataset Upload
- Model Training
- Churn Prediction
- Customer Recommendations
- Dashboard Analytics

Include validation, pagination, filtering, and Swagger/OpenAPI documentation.

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11. Logging & Monitoring

Track:

- Model Training History
- Prediction Requests
- API Performance
- Error Logs
- User Activity

Bonus Features

- AutoML Integration
- SHAP Explainability Dashboard
- Scheduled Model Retraining
- Real-Time Prediction API
- Email Alerts for High-Risk Customers
- Export Reports (PDF/CSV)
- Docker Deployment
- CI/CD Pipeline

Deliverables

The intern must submit:

- GitHub Repository
- Complete Source Code
- README Documentation
- API Documentation
- Sample Dataset
- Trained Model Files
- Docker Configuration
- Postman Collection
- Deployment Link

Evaluation Criteria

Criteria	Weight
Data Preprocessing & EDA	20%
Machine Learning Model Development	25%
Customer Segmentation & Recommendation	20%

Backend APIs	10%
Frontend Dashboard	10%
Code Quality & Documentation	10%
Bonus Features	5%

Final Presentation

Prepare a **10–15 minute project demonstration video** covering:

- Problem Statement
- Dataset Overview
- Data Cleaning & Feature Engineering
- Customer Segmentation
- Model Training & Evaluation
- Churn Prediction Workflow
- Recommendation Engine
- Dashboard Demonstration
- API Testing
- Deployment Process
- Challenges Faced
- Future Improvements